EC571 Project Report

A 28 nm Configurable Memory (TCAM/BCAM/SRAM) Using Push-Rule 6T Bit Cell Enabling Logic-in-Memory

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1 Introduction

Traditionally, in memory architecture, Content addressable memory (CAM) carries out parallel search operations, while the SRAM performs reading and writing. Binary CAM (BCAM) cell consists of 10 transistors (6T SRAM and 4T comparison logic). This occupies lots of space when scaled. In order to target this problem, the paper [1] proposed an organization that operates the traditional 6T SRAM array in a way such that search operations or logic in memory operations are supported. This design would increase the CAM density by reducing the transistor count and allowing push-rule to be used for manufacturing. In this lab, we implemented an example of this proposed memory array and simulated its functionality. In all our simulations, the load capacitance is 100fF, and VDD is 1V unless otherwise specified.

2 Memory Cell

The proposed BCAM cell is chosen to be the same as a standard SRAM cell to make the manufacturing process more efficient. The schematic for this cell is shown in Figure 1. The access NMOS

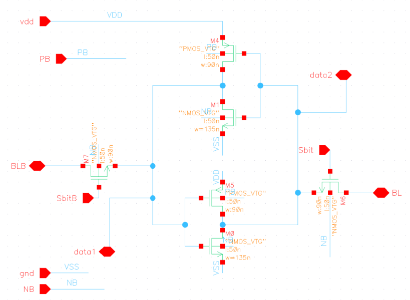


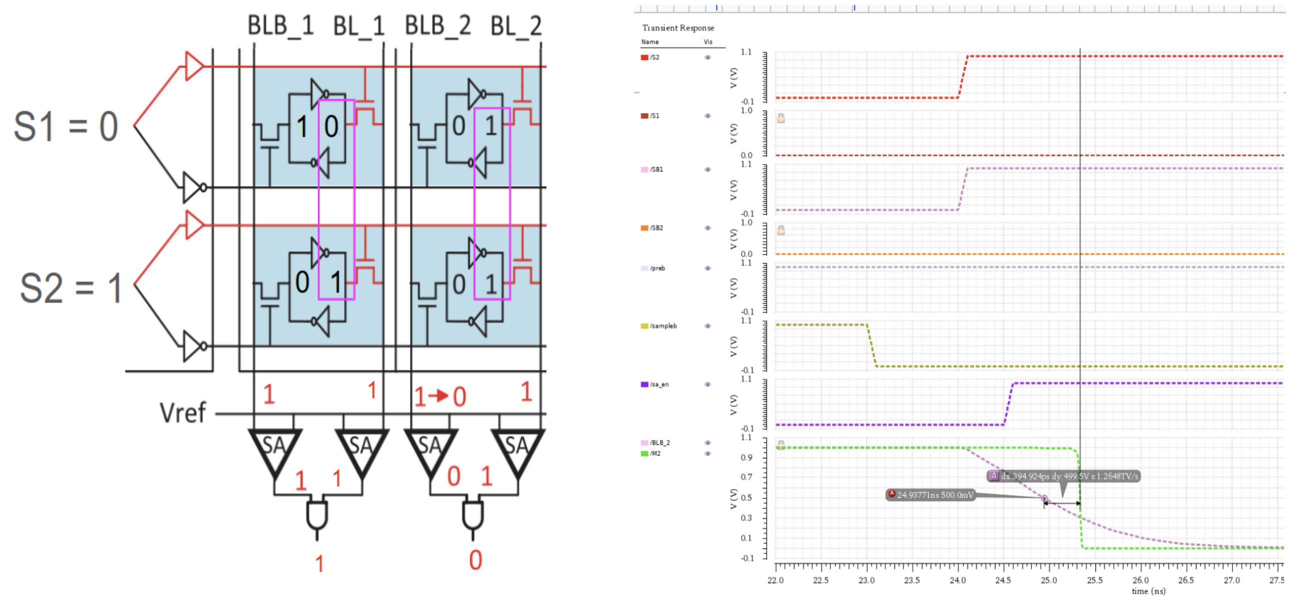
Figure 1: Schematics for Proposed BCAM cell

transistors have a minimum width of 90nm, which is smaller than the pull-down NMOS (135nm) to prevent read upset. To ensure that the pull-up is weaker than the access to allow write, the PMOS also has a minimum size of 90nm.

3 Memory Array

3.1 Search Operations

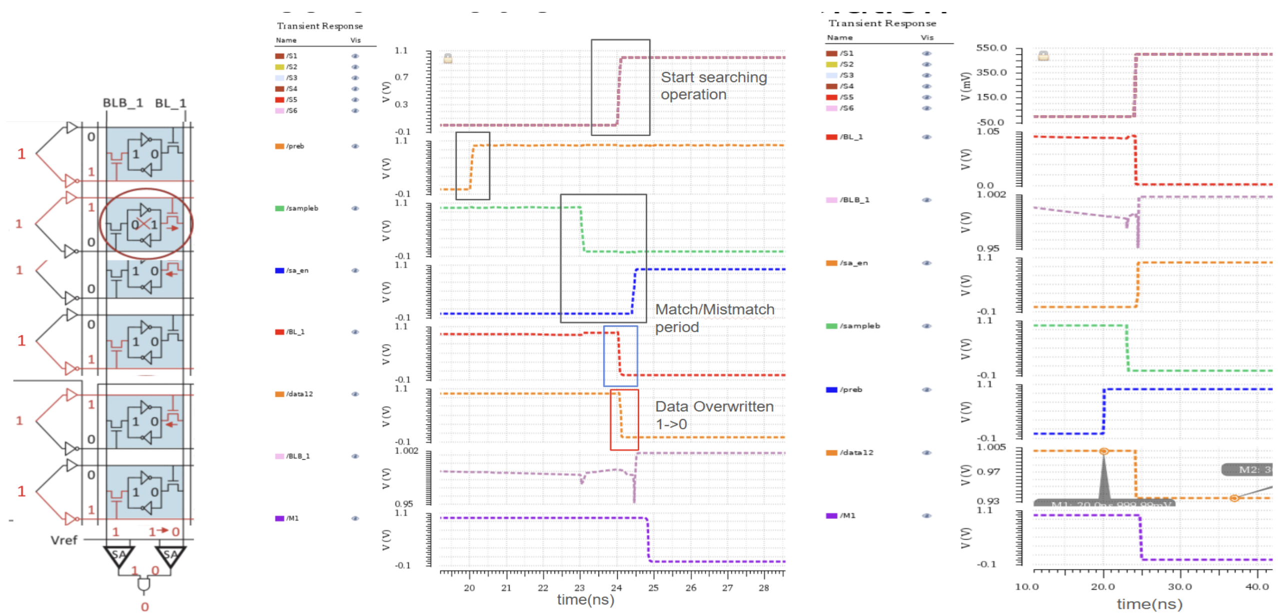
Searching operation works the similar with standard SRAM read. In SRAM read, the two access transistors are enabled by the wordline. Whilst the SRAM search has an access transistor controlled by a search data and the other controlled by the $\overline{\text{search data}}$. Bit line (BL) and $\overline{\text{Bit Line}}$ are precharged to VDD. During the search operation, a mismatch would be detected when the search data is logic 1, and the stored data has logic 0. The bit lines would not discharge when search data is logic 0 and stored data is logic 1. The complement of search data and stored data helps to detect the mismatch. The mismatches are detected with either of the bit lines discharged using AND gates. To improve search speed, the sense amplifier detects the mismatch when bit lines are lower than 0.8V.



(a) Searching with a match and a mismatch (b) Mismatch (M2) starts pulling down at approximately 0.8 V

Figure 2: Search Operations

3.1.1 Search Problem: Overwritten stored data



(a) Overwritten (b) Before configuration (c) After configuration

Figure 3: Search Operations' solution

During searching for a memory address, the memory array may have multiple mismatches, discharging the bit lines at a faster rate. When bit lines fall below the voltage threshold while still in search operation, the matched data with stored logic 1 may be discharged through bit lines. This leads to stored logic being overwritten and results in corrupting memory.

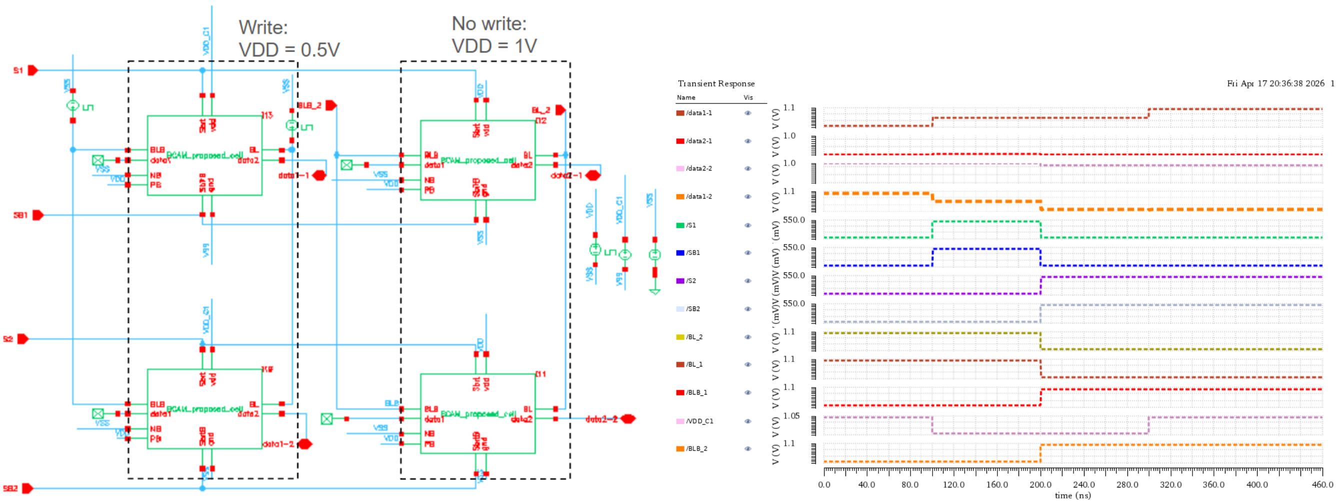
In order to solve this problem, two different power lines are used. Coupled inverter and stored logic 1 are driven by a lower VDD (VDD_Lo), while the Bit lines remain precharged to full VDD. According to experiments in [1], the optimal VDD_Lo is half the VDD. In our following simulations, VDD is chosen to be 1V and VDD_Lo to be 0.5V. Since the threshold voltage for our devices is 0.335V, the stored logic 1 could only be overwritten when the bit lines' voltage falls below 0.165 V. However, since sense amplifier sampling logic 0 always happens when bit lines fall below 0.8 V, quickly precharge bit lines to 1 V. This helps to avoid overwriting the data.

3.2 Write Operations

Writing into a selected row of the memory array follows the same procedure as standard SRAM writes. However, since the data is stored in columns in this proposed organization, writing should be performed column-wise. To achieve this, two clock cycles have to be used: in the first cycle, the word line is set to high for all the rows that should store a 1; in the second cycle, the word line is set to high for all the rows that should store a 0. To write in, in the first cycle, the bitline is set to high and the bitline-bar to low so that their voltage can pass to the inverter latch through the two access transistors that are turned on. In the second cycle, to write a zero, the bitline is low, and the bitline-bar is high.

To prevent this from overwriting the unselected columns, a write-assist technique is employed. The wordlines and the supply voltage for the column to be written are set to a lower VDD (VDD_Lo), while the other columns are supplied with the same full VDD. Due to the pass-transistor characteristics, this under-driven wordline can only pass voltages below VDD_Lo, which is interpreted as a zero in the full-VDD cells, making it impossible to flip bits. VDD_Lo is 0.5V, same as the previous search problem solution.

We simulated this with the testbench shown in Figure 4a. The first column (initially 01) is to be written with the string 10, and the second column should stay the same as its initial value (01). The resulting waveform is shown in Figure 4b, demonstrating a successful write.



(a) Testbench Schematics

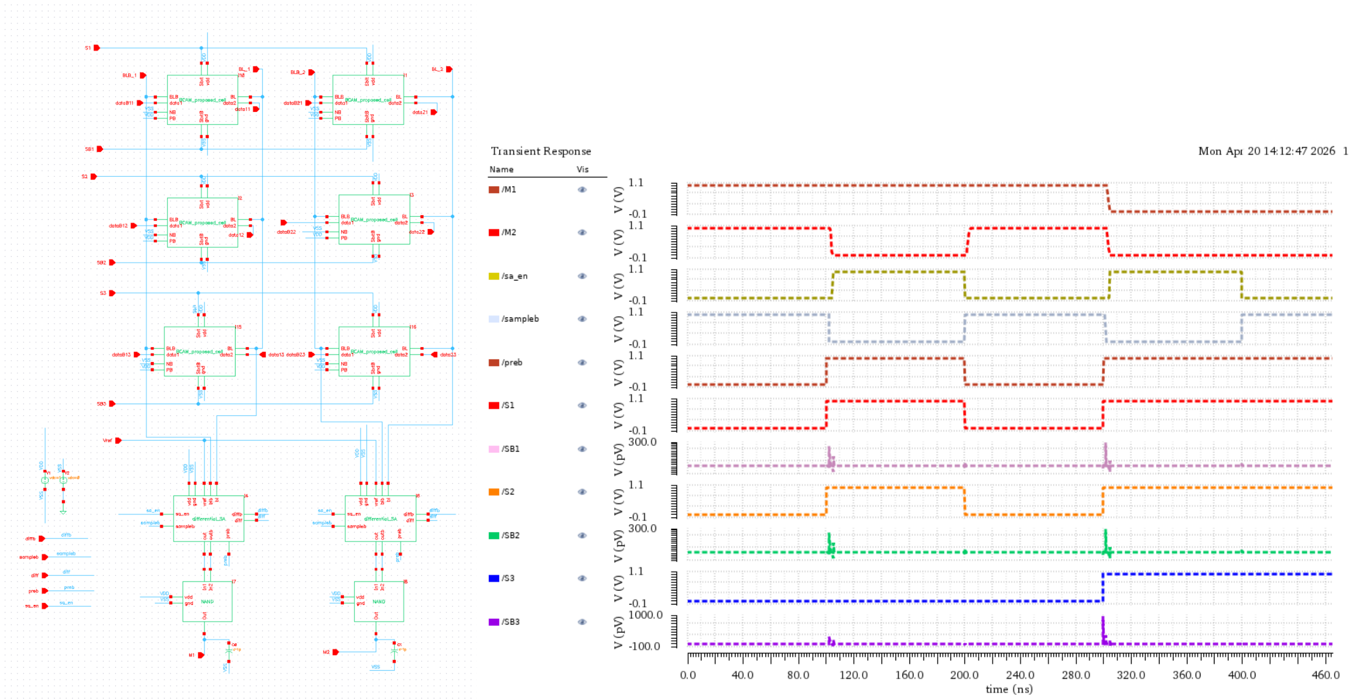
(b) Waveform

Figure 4: Simulation Results of Column-wise Write Operations

3.3 Logic in Memory (LIM) Operations

After the bitline is precharged to high, it will discharge if and only if one or more of the cells are 0, which is equivalent to an AND operation. Therefore, an AND can be performed between two or more rows by setting the wordline of the chosen rows to high and keeping the wordline-bar and other wordlines at low.

We simulated this behavior with the testbench shown in Figure 5a that consists of two columns with initial data 110 and 001, respectively. The simulated waveform is shown in Figure 5b. In the first cycle, the first two rows were ANDed, and the third row was masked (wordline and wordline-bar set to low). The first column output (M1) was high, and the second column output (M2) was low, as expected. During the second cycle, all three rows have wordlines set to high, and a 3-input AND was performed. As demonstrated in the waveform, M1 and M2 both dropped to low, indicating a correct calculation.



(a) Testbench Schematics

(b) Waveform

Figure 5: Simulation Results for AND Opeartion

Apart from AND, NOR operation can also be performed by ANDing all the complements of the value stored, according to De Morgan's law. To achieve this, instead of the wordlines, we set the wordline-bars to high to access the data-bars. The remaining procedure is the same as the AND function, so we will skip the demonstration.

4 Configurable Sense Amplifier

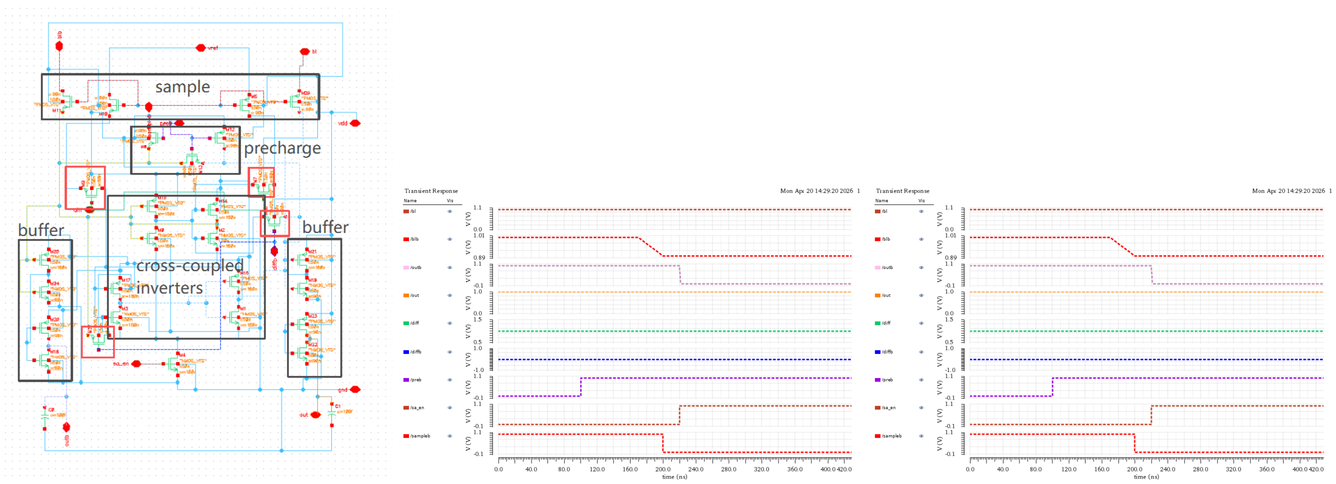
Since the proposed cell can be used both as an SRAM and a BCAM, the sense amplifier must support both configurations, including SRAM read, SRAM LIM, and BCAM search. The sense amplifier schematic is shown in Figure 6a.

The sampling PMOS transistors and precharge PMOS transistors are designed with a minimum width of 90nm to minimize area. The first inverter should be high-skew, and the second should be low-skew because the final output is monotonically falling. Two pairs of cross-coupled inverters compare and amplify the voltage difference on their two sides. Notice that the pull-down network consists of an NMOS in the inverter pair, and an enable NMOS that controls when the sense amplifier is on. They are all sized twice the minimum width (180nm) to match the resistance of a standard unskewed inverter. The transistors boxed in red control what the cross-coupled inverters compare based on the complementary *diff* and *diffb* signals. These are of a minimum size of 90nm.

We observed that the search operation delay is longer than that of conventional, and a great portion comes from the sense amplifier. After simulating through different sizing for buffer and coupled inverters, we found that increasing the buffer size according to the output capacitance significantly improves the detection speed. Since there is only one sense amplifier for each column that consists of many cells, this increase in width does not considerably affect the overall area. In the case of optimizing the area, the cross-coupled inverter can be scaled to a minimum size (90nm) to compensate without noticeably compromising performance.

4.1 BCAM Mode

The sense amplifier is in BCAM mode when *diff* is low and *diffb* is high. If any of the bitlines drop below V_{REF}, then the inverters would amplify this difference into a full VDD swing. To ensure robustness against noise while having a faster amplification speed, the reference voltage is chosen to be 0.7V when VDD is 1V. In this simulation, the bitline-bar is set to drop from 1V to 0.6V to emulate a search mismatch, and the bit-line is set to 1V throughout to emulate a search match. The sense amplifier starts sampling after the voltage change on the inputs occurs to allow this change to develop across the cross-coupled inverter pairs. Then the amplifier is turned on after that by setting *sa_en* to high. The simulation waveform is shown in Figure 6b, demonstrating a correctly amplified signal on the bitline-bar.



(a) Schematics (b) Waveform for BCAM Mode (c) Waveform for SRAM Mode

Figure 6: Schematics and Simulation Results for Sense Amplifier

4.2 SRAM Mode

The sense amplifier is in SRAM mode when *diffb* is low and *diff* is high, comparing between the bitlines instead of against a reference voltage. The control signal timings are the same as the previous BCAM mode. The resulting waveform is shown in Figure 6c.

References

[1] S. Jeloka, N. B. Akesh, D. Sylvester, and D. Blaauw, "A 28 nm configurable memory (tcam/bcam/sram) using push-rule 6t bit cell enabling logic-in-memory," *IEEE Journal of Solid-State Circuits*, vol. 51, no. 4, pp. 1009–1021, 2016. DOI: 10.1109/JSSC.2016.2515510.